Evidence appendix to a BlueQubit Quantum Flywheel RFP proposal, Track 2

Adversarial Replication Flywheel

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About this appendix. This is the evidence appendix for a proposal to automate classical attacks on quantum-advantage claims: searching for cheaper ways to reproduce quantum-computer output on ordinary hardware. It reports one *preregistered* campaign: rules, thresholds and predictions were fixed before the first job ran, as in a clinical trial. A self-improving attack pipeline (the *flywheel*) competed for 10 rounds against a *no-memory control* with the same job budget on a frozen 48-circuit suite. A code-enforced gate, never a person, decided every promotion; both finals were then scored on a sealed 12-circuit *holdout*, a test set touched exactly once. The score is *XEB fidelity* (cross-entropy benchmark): how closely the simulator's samples match the circuit's ideal output (1 = exact; higher = better classical mimicry of the quantum device). Every fidelity number resolves to a BlueQubit job ID in the experiment ledger (located in Table 3). Stated plainly: both arms beat the baseline, but the flywheel matched the control and did not beat it (holdout difference +0.0017 XEB, 95% CI [−0.0058, +0.0098] includes zero).

1/10 vs 1/10
flywheel vs control promotions (10 preregistered rounds each arm)

+0.0479
control holdout gain over baseline r0, 95% CI [+0.0100, +0.0886]

$0.00
total spend for 865 ledgered jobs on CPU-class devices (865 mps.cpu); GPU/QPU never used

What ran. The flywheel proposes one simulation-recipe mutation per round (MPS bond-dimension allocation χ, qubit permutation, or shot allocation), scores candidate against incumbent on 12 validation circuits, and promotes only on a statistically significant fidelity gain at non-inferior cost. The control runs the identical harness, job budget and gate with a fixed proposal ladder while retaining its promoted incumbent — cross-round memory is disabled, so the comparison with the adaptive arm is descriptive, not a pure memory-only ablation. Both arms completed all 10 rounds; the frozen 12-circuit holdout was unlocked and scored a single time after the campaign closed, then permanently resealed by filesystem sentinels.

Gate (identical for both arms, enforced by the ledger — never by hand): paired 95% CI on mean ΔXEB excludes 0 AND mean Δ > min_delta = 0.02544 (= 2× measured run-to-run sd) AND cost non-inferior within 2%. Score = mean per-circuit normalized XEB (1 = exact). Job plan per round: 24 train + 12 val, ≤2 concurrent, 15-minute wall cap with ≥3× headroom.

Holdout result — evaluated once, 12 circuits, 36 jobs

Table 1. One-shot holdout (G6). Flywheel − control paired: +0.00170, CI [−0.00580, +0.00979].

configuration	mean holdout XEB	paired Δ vs r0	95% CI
baseline r0 (replicate)	0.92652	ref.	ref.
no-memory control final	0.97441	+0.04789	[+0.01002, +0.08863]
flywheel final	0.97611	+0.04959	[+0.01375, +0.08890]

Headline. The flywheel's round-1 χ-allocation mutation (r1, chi_alloc = easy32hard128, Δ +0.03800, CI [+0.00243, +0.07996] — excludes 0) cleared every gate clause and was promoted; the no-memory control, running its fixed proposal ladder, promoted the identical recipe at its own round 1. On the locked holdout both finals confirmed genuine, locked-in gains over the baseline replicate (+0.0496 / +0.0479, CIs exclude 0) at equal-or-better wall time; flywheel vs control stayed +0.0017 (CI [−0.0058, +0.0098] includes 0) — no measurable memory edge over the tuned control at this scale, reported as such. All nine later mutations (χ variants, permutations, shot reallocations) were correctly rejected.

Gate calibration (EXP-008). Fidelity-component pass probability 58.5% at the observed effect (pooled mean paired Δ +0.03389), 33.5% at a true effect sitting exactly at min_delta; worst same-config rep noise 0.49× min_delta.

- Design: 48 fresh circuits (seed 20261007, zero suite/holdout contact), baseline uniform64 vs challenger easy32hard128 (easy d≤8 / hard d≥12), 20000 shots × 2 reps — 192 jobs on mps.cpu, TEST account, Oct 7 ($0.00).
- A1/A2/A3: pooled paired Δ +0.03389 (per-rep +0.03632/+0.03146); bootstrap 95% CI [+0.01745, +0.05212]; rep noise |rep1−rep2| = 0.49× (base) / 0.47× (chal) of min_delta 0.026952.
- A4/A5: at the 12-circuit decision scale P(both) = 58.5% observed / 33.5% shifted-to-min_delta; per-rep single-evaluation libraries 66.1% / 45.9%. Fidelity components only — the cost gate (ratio ≤ 1.02) is not simulated (deterministic per config). Frozen prediction (point +0.045, band [+0.015, +0.080]): observed +0.03389, in_band = True.

Round-by-round staircase

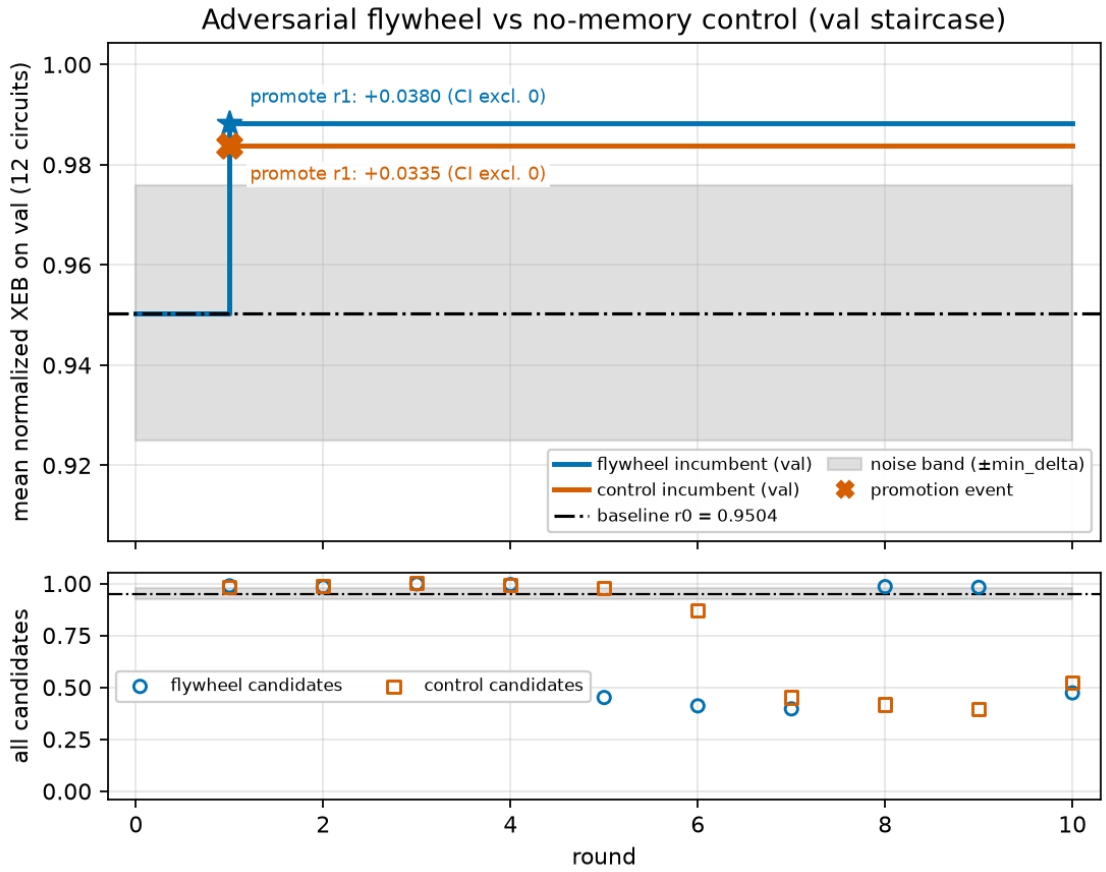


Fig. 1. Incumbent trajectory per round (top, zoomed to the noise band) and every candidate score (bottom). Blue = flywheel, orange = control: each arm promoted its round-1 χ-allocation (Δ +0.0380 / +0.0335, CIs exclude 0) then stayed flat for the remaining nine rounds — no later mutation cleared the gate. The control's promoted config costs 28.8 vs 33.7 min on validation (15% lower val cost). The gray band is ±min_delta.

Table 2. All 20 rounds, as recorded by the ledger (cost ratio = candidate / incumbent cost); alternate rounds are shaded.

arm	r	mutation	value	Δ mean	95% CI	cost	verdict
flywheel	1	chi_alloc	easy32hard128	+0.03800	[+0.00243, +0.07996]	0.861	**PROMOTED**
control	1	chi_alloc	easy32hard128	+0.03352	[+0.00116, +0.06997]	0.854	**PROMOTED**
flywheel	2	chi_alloc	uniform128	−0.00456	[−0.01458, +0.00639]	1.179	rejected
control	2	chi_alloc	uniform128	+0.00312	[−0.00844, +0.01384]	1.181	rejected
flywheel	3	chi_alloc	easy32hard256	+0.00994	[−0.00281, +0.02321]	1.015	rejected
control	3	chi_alloc	easy32hard256	+0.01911	[+0.00413, +0.03419]	1.041	rejected
flywheel	4	chi_alloc	uniform256	+0.00633	[−0.00579, +0.01954]	1.230	rejected
control	4	chi_alloc	uniform256	+0.01074	[−0.00343, +0.02542]	1.259	rejected
flywheel	5	permutation	11	−0.53729	[−0.68423, −0.35685]	1.151	rejected
control	5	chi_alloc	easy16hard128	−0.00610	[−0.02105, +0.00814]	1.008	rejected
flywheel	6	permutation	23	−0.57739	[−0.69914, −0.44737]	1.137	rejected
control	6	chi_alloc	uniform32	−0.11167	[−0.20534, −0.02947]	0.869	rejected
flywheel	7	permutation	37	−0.59192	[−0.69883, −0.47639]	1.155	rejected
control	7	permutation	11	−0.53231	[−0.67533, −0.35297]	1.171	rejected
flywheel	8	shots_alloc	easy10khard30k	−0.00368	[−0.01880, +0.01179]	0.997	rejected
control	8	permutation	23	−0.56659	[−0.68559, −0.43889]	1.147	rejected
flywheel	9	shots_alloc	easy4khard36k	−0.00681	[−0.01748, +0.00374]	0.986	rejected
control	9	permutation	37	−0.58823	[−0.69197, −0.47538]	1.148	rejected
flywheel	10	permutation	211	−0.51502	[−0.65080, −0.36211]	1.153	rejected
control	10	permutation	51	−0.46232	[−0.61072, −0.30383]	1.094	rejected

Cost–fidelity Pareto and the χ mechanism

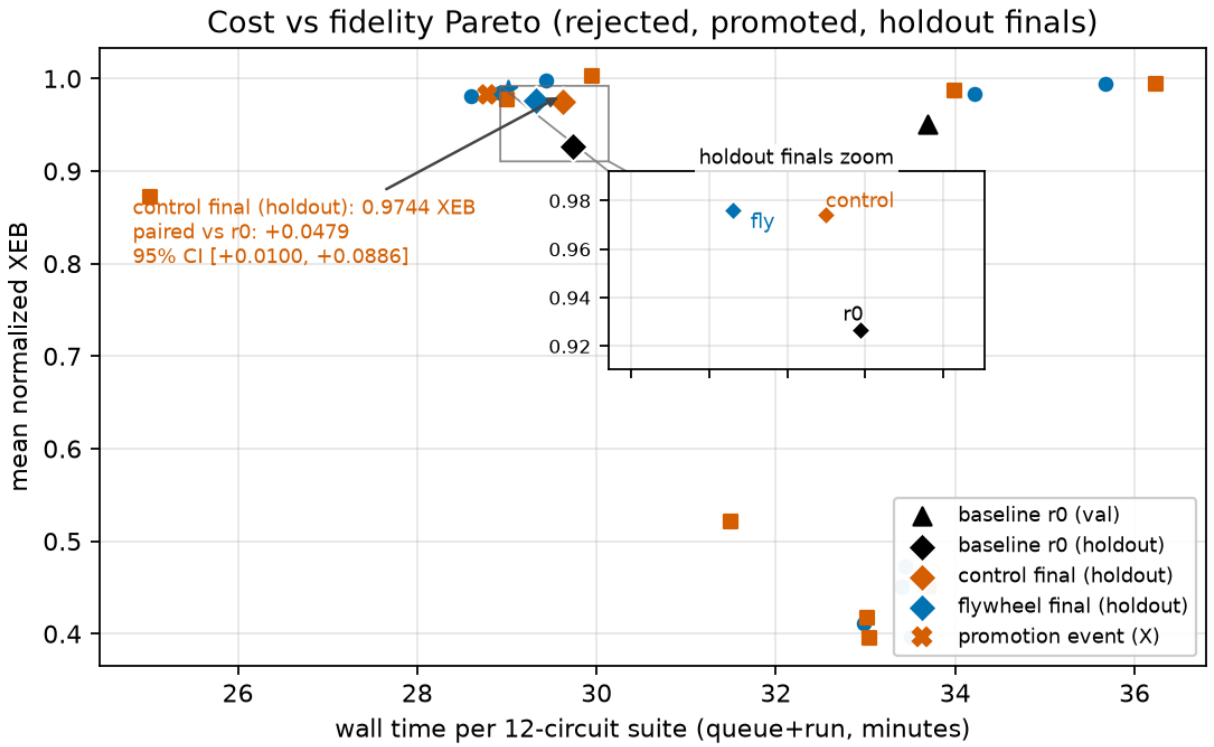


Fig. 2. Cost (wall minutes per 12-circuit split) vs fidelity; validation-candidate points and holdout finals sit on different suites and are read separately. Both holdout finals dominate the baseline replicate (higher XEB at equal-or-better wall time); the flywheel final (blue) tops the holdout finals, +0.0017 above the control final (orange diamond) — a CI-includes-0 difference, read as parity, not superiority.

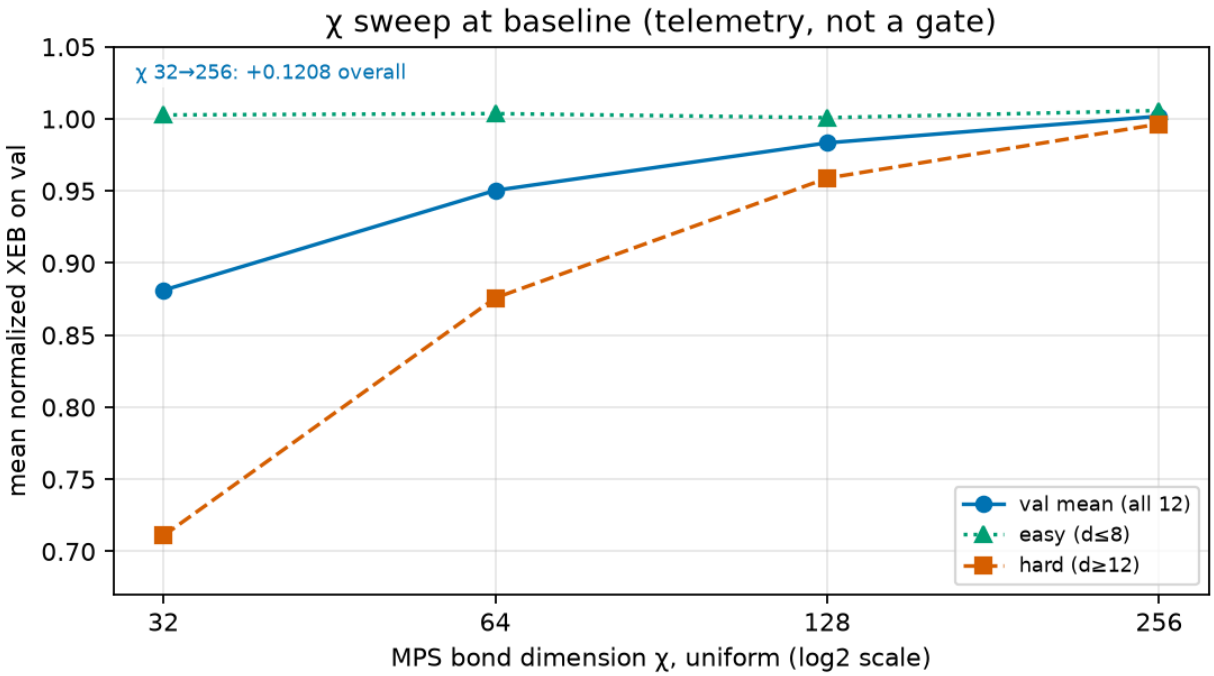


Fig. 3. χ telemetry at baseline: hard circuits (dashed) are χ-starved — low bond dimension destroys them; easy circuits (dotted) are not. This is the mechanism behind the r1 promotions in both arms (shifting χ budget to hard circuits) and the permutation crashes: a permuted layout changes which circuits are hard.

Honest negatives — reported as designed outcomes

No measurable memory edge. 1/10 flywheel promotions vs 1/10 control — both arms promoted the identical round-1 χ reallocation (the control found it via its fixed ladder, the flywheel via adaptive search). On the locked holdout the flywheel final leads the control final by +0.0017 XEB with CI [−0.0058, +0.0098] including 0 and holdout wall-time equal or better: 29.3 vs 29.6 min: at this scale the memory matched a tuned no-memory control and did not beat it, and this appendix reports that as the result.

The frozen prediction HIT. Point 0.97519, band [0.95038, 0.99900], actual 0.97611 — in_band = True. The suite was frozen at gate G2; the prediction was frozen at 2026-10-11T19:33:14+00:00 — immediately after the G3 baseline, before any mutation round was dispatched — and scored exactly once at 2026-10-12T15:25:59+00:00. The band, set from validation noise before holdout scoring, contains the actual with room to spare; the predict→score discipline held and the model was right this time.

χ-dependent permutation damage. Permutation mutations destroyed fidelity across all eight permutation rounds (Δ −0.46 to −0.59) under low-χ allocation: classical layout heuristics are fragile exactly in the regime adversarial simulation must probe.

A routing incident was disclosed, not hidden. The 36 holdout-scoring jobs were dispatched to the TEST account by an orchestrator environment omission (all 829 campaign jobs ran on the personal account; the ledger records device and cost, not account, and mps.cpu results are account-independent). The run was left to finish rather than killed — the one-shot holdout sentinel was already unlocked — and the omission is disclosed here and in the campaign log instead of being patched in silence.

Known limitations

- Splits are not depth-stratified: the holdout has 4× depth-4 circuits, val none, so the ≈0.024 val→holdout gap partly reflects composition.
- The cost gate is a point-ratio screen on noisy wall-times from a single device class (mps.cpu).
- Shot presets assumed 6 easy / 6 hard circuits; the suite manifest is 7/5 (220k/208k vs 240k shots).
- All confidence intervals are pointwise 95%; there is no multiplicity control across the 20 rounds.
- The identical round-1 recipe was promoted in both arms, so the holdout cannot separate adaptive search from the fixed ladder on this suite — only the nine rejected follow-up rounds differ.

Traceability and reproducibility

Every performance claim carries job IDs. Exact reference probabilities are frozen local statevector files; the two derived job-less claims (threshold formula, frozen prediction) cite their formulas and the frozen prediction record (researcher/PREDICTIONS.jsonl). The report ships a machine-checked claims ledger: 31 claims referencing 432 jobs (336 distinct), all resolving against the 865-job immutable ledger (ledger/jobs.sqlite; exports in runs/report/jobs_export.{json,csv}). The holdout suite is frozen (sha256 2e583cf8…), locked mode 0 with sentinels and an access log; with the .HOLDOUT_EVALUATED tombstone in place, a second evaluation is blocked by the harness — the ledger cannot record a job outside a REGISTERED experiment.

The funded system starts from this code — already re-run end-to-end. The same harness executed the canonical campaign a second time on the personal account with zero code changes from the original TEST campaign (OCT11-RUNBOOK.md); this appendix documents that rerun. The grant scales the suite and the mutation families to the 3-month plan, not the $0 rig — the ledger, gates, frozen holdout and one-shot discipline carry over unchanged.

Table 3. Evidence manifest. Total spend: $0.00; all 865 ledgered jobs on CPU-class devices (865 mps.cpu); GPU/QPU never touched. The 829 campaign jobs ran on the personal account, the 36 holdout-scoring jobs on the TEST account (the disclosed routing incident above).

artifact	where
campaign report + figures + claims ledger	runs/report/ (report.md, claims.json, staircase.png, pareto.png, chi_sweep.png)
per-round outcomes	runs/report/rounds.json + ledger/jobs.sqlite (865 jobs)
preregistered gate + frozen prediction	researcher/PREDICTIONS.jsonl (frozen at G3, scored once at G6)
sealed holdout + access audit	suite/ (mode 0, sentinels) + HOLDOUT_ACCESS.log
campaign procedure (rerun of record)	OCT11-RUNBOOK.md